

PROOF OF CONCEPT | COMPANY A WIRELESS TELECOM

Smart Plan Migration

Transforming Telecom 'Bill-Shock' into Recurring Premium Revenue via Predictive ML Upselling

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AGENDA

I. Analysis & Modeling

- Current Market Landscape & Problem Statement: Traditional churn vs. proactive upselling.
- Exploratory Data Analysis & Target Identification: The overage structure.
- Machine Learning Framework: Comparing baseline and advanced models.
- Validity & Precision-Recall Balancing: Identifying the right threshold.

II. Business Proposal & Financials

- Overage Upsell Proposal: Campaign design, targeting logic, and win-win migration.
- Projected Financial Impact: Waterfall chart of ARPU lift and saved CLV.
- Risk Assessment & Sensitivity Analysis: ROI resilience testing.
- Commercial Quotation & Partnership: One-time system licensing and ongoing support.
- Implementation Roadmap: Structured 3-phase rollout plan.

MARKET CONTEXT: THE COST OF BILL-SHOCK

Telecom Industry Context

- Overage Pain Point: High overages are a primary source of customer friction, resulting in customer anger and sudden churn.
- Market Shifts: With competitive flat-rate and unlimited options, customers demand predictable monthly charges.
- Traditional Churn Limits: Standard churn models are reactive. By the time a customer is identified, they are already dissatisfied. Preempting this with upsell plans preserves margins.

1.9x

CHURN RISK MULTIPLIER

Overage candidates churn at nearly double the rate of regular subscribers

57.1%

OVERAGE PENETRATION

Over half of the active customer base suffers monthly bill-shock

HYPOTHESIS & EDA

The Target Hypothesis

- Rather than waiting for churn, we target customers on the WRONG plans.
- Upsell Candidate definition:
 - Monthly overage $\geq$ 15% of total bill, OR
 - Absolute overage $\geq$ \$29.20 (top quartile of overage payers).
- This cohort accounts for 34.6% of the customer base.
- Migrating them to premium plans is a value-creating, margin-positive play.

34.6
%

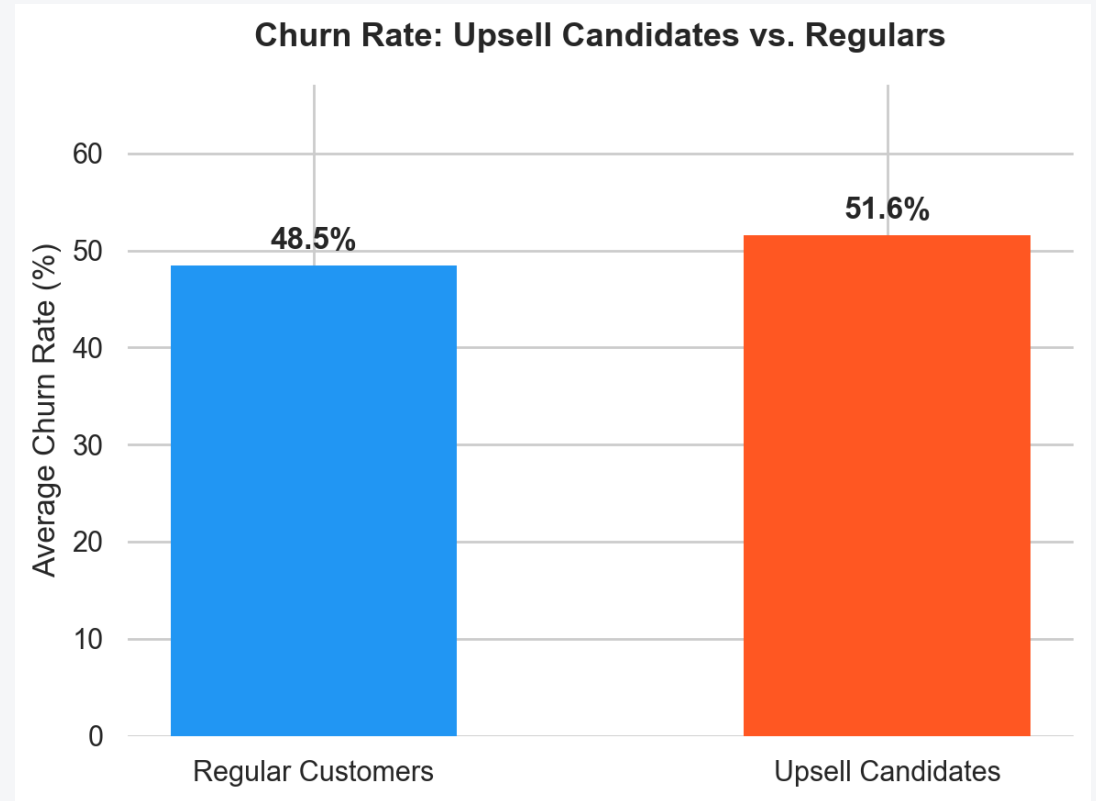
COHORT SIZE

Subscribers identified as upsell candidates

\$23.6
7

AVG OVERAGE FEE

Mean monthly penalty paid by mismatched users



DATA ENGINEERING & PREPROCESSING

Dataset Merging & Imputation

- Data Ingestion: Client demographics (Client.csv) and usage history (Record.csv) were inner-joined on Customer_ID (100k matched rows).
- Imputation: Numeric nulls filled with median values; categorical nulls filled with mode values. Columns with high missingness (e.g. numbcars, dwllsize) imputed to retain context.
- Encoding: Categorical columns (area, marital status, credit availability) converted via LabelEncoding for the tree-based pipeline.

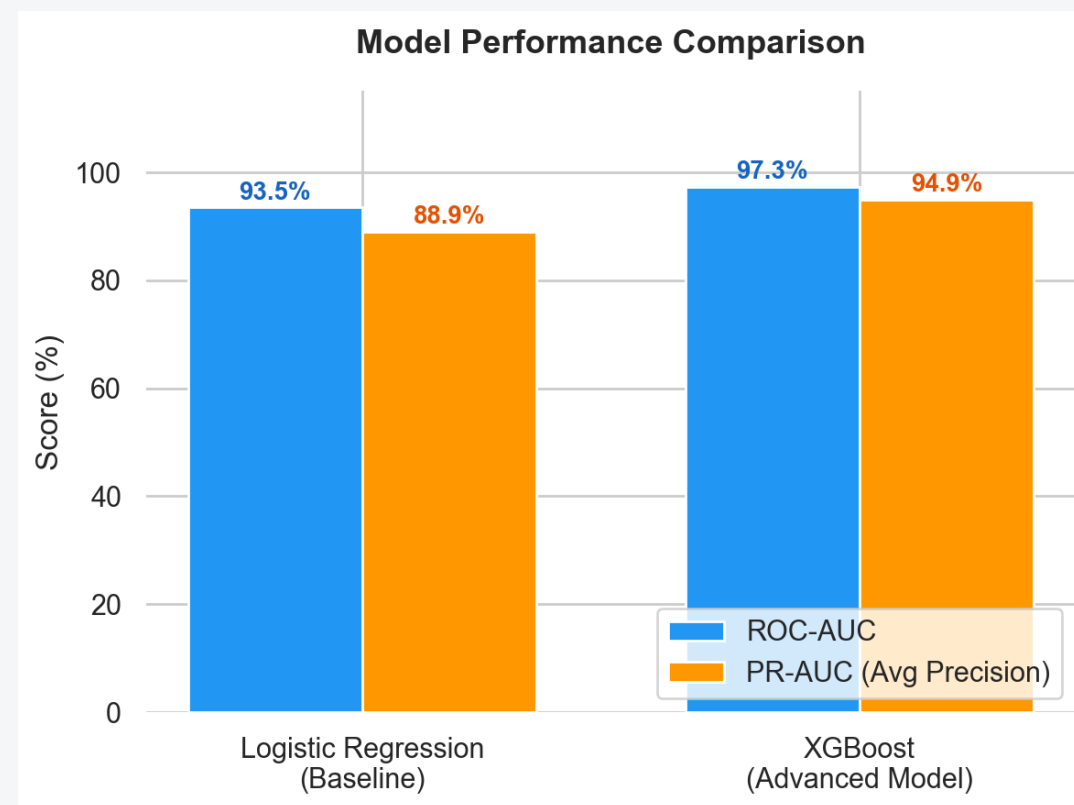
Custom Feature Engineering

- Overage-to-Revenue Ratio: Measures target segment alignment. Excluded during training to prevent target leakage.
- Usage Dropoff Rate: Captures percentage change in minutes of use ($\text{change_mou} / \text{mou_Mean}$). Highlights disengaging customers vs. growing usage profiles.
- Revenue Per MOU: Mean revenue divided by monthly usage minutes. Serves as a plan-efficiency benchmark.

MODEL COMPARISON: BASELINE VS. ADVANCED

Evaluating Modeling Frameworks

- **Baseline Model:** Logistic Regression with StandardScaler. Achieves a strong ROC-AUC of 93.5% and PR-AUC of 88.9%.
- **Advanced Model:** XGBoost Classifier. Leverages non-linear feature interactions to boost performance, reaching an outstanding ****97.3% ROC-AUC**** and ****94.9% PR-AUC****.
- **Model Selection:** XGBoost is chosen as the Champion Model. The +3.8% ROC-AUC and +6.0% Average Precision gains translate directly to thousands of correctly identified targets, maximizing campaign efficiency.



OVERVIEW OF THE CHAMPION MODEL

96

INPUT FEATURES

Predictors utilized in modeling

1.89

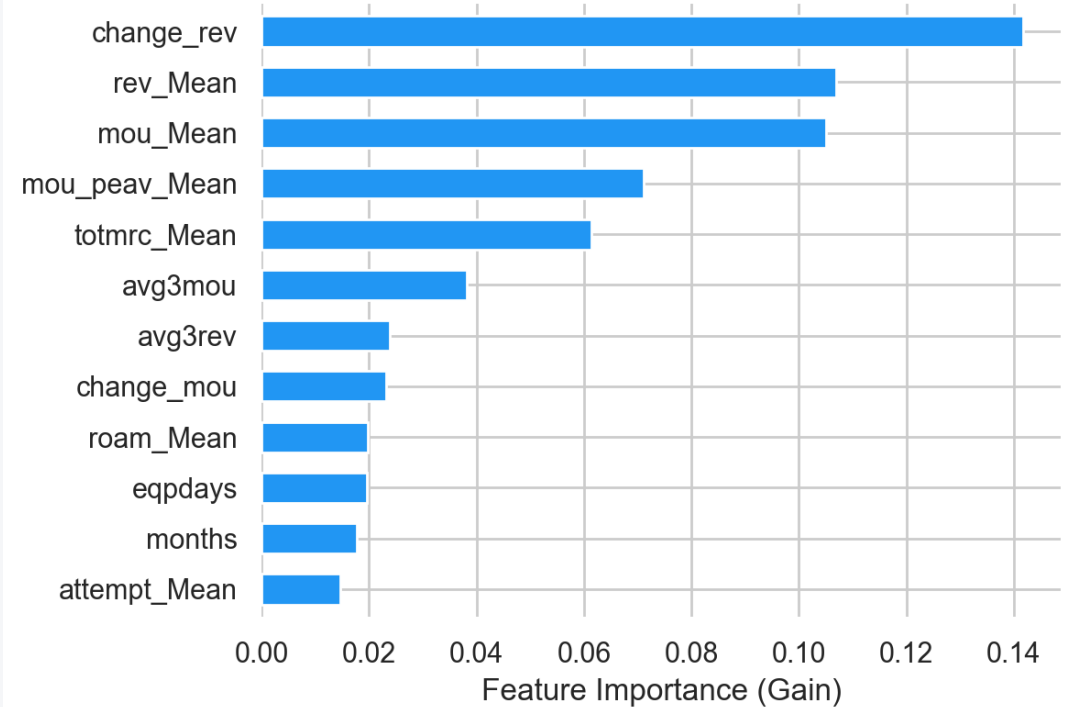
POS WEIGHT ADJ

Class weight adjustment for imbalance

XGBoost Configuration

- Algorithm: XGBoost Ensemble Classifier (100 base trees, learning rate 0.1, max depth 5).
- Features: 96 predictors spanning billing history, demographics, and drop call patterns.
- Data Leakage Prevention: Excluded direct overage features (ovrrev, ovrrou, datovr, vceovr) from training to ensure validity on unseen data.

Top Drivers of Plan Mismatch (XGBoost)



CHAMPION MODEL VALIDITY & PERFORMANCE

97.3%

MODEL ROC-AUC
Excellent class separation

94.9%

AVERAGE PRECISION
Very low false positive rate

Model Evaluation Metrics

- Overall Test Set Accuracy: 92.0%
- Target Class Precision: 86.0% (minimizes marketing spend waste)
- Target Class Recall: 93.0% (captures the vast majority of target cohort)

Model Validation Results (Test Set, N = 20,000)

Class Label	Pred. Regular	Pred. Upsell
True Regular	11,888 (92%)	1,189 (8%)
True Upsell	502 (7%)	6,421 (93%)

- Model achieves a low False Positive rate (8%), which is critical to avoid contacting customers who are already on the correct plan.
- High recall (93%) ensures we capture almost all potential upsell revenue.

REVENUE MAXIMIZATION PROPOSAL

**¥298.2
M**

**NET ANNUAL PROFIT
LIFT**

Annual ARPU lift + Churn-save
value minus campaign contact cost

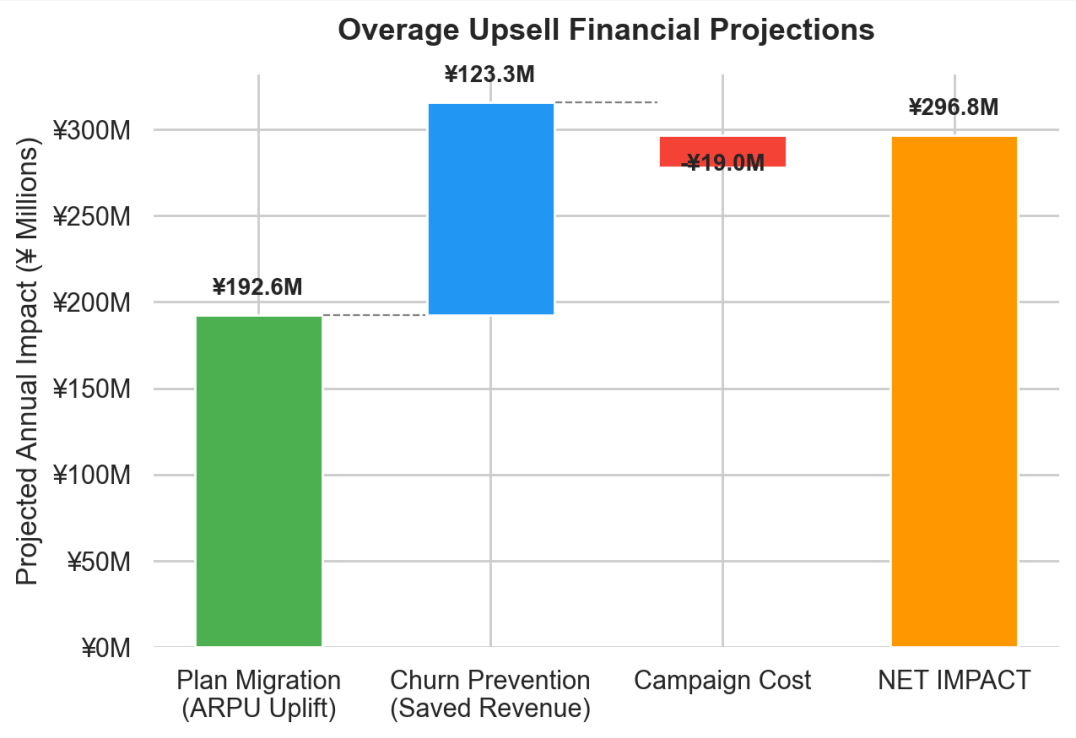
1,586%

CAMPAIGN ROI

Exceptional return due to highly
targeted outreach

Proposal Parameters & Campaign Logic

- Migration uplift: Assuming 10% conversion to Premium Plan (+¥5,000/mo), generating ¥193.3 Million annually.
- Saved Churn: Retaining 20% of converted customers who would have otherwise churned, saving ¥123.6 Million in Lifetime Value.



SENSITIVITY ANALYSIS & RISK CONTROLS

1.5%

BREAK-EVEN CONVERSION

Extremely low threshold for
profitability

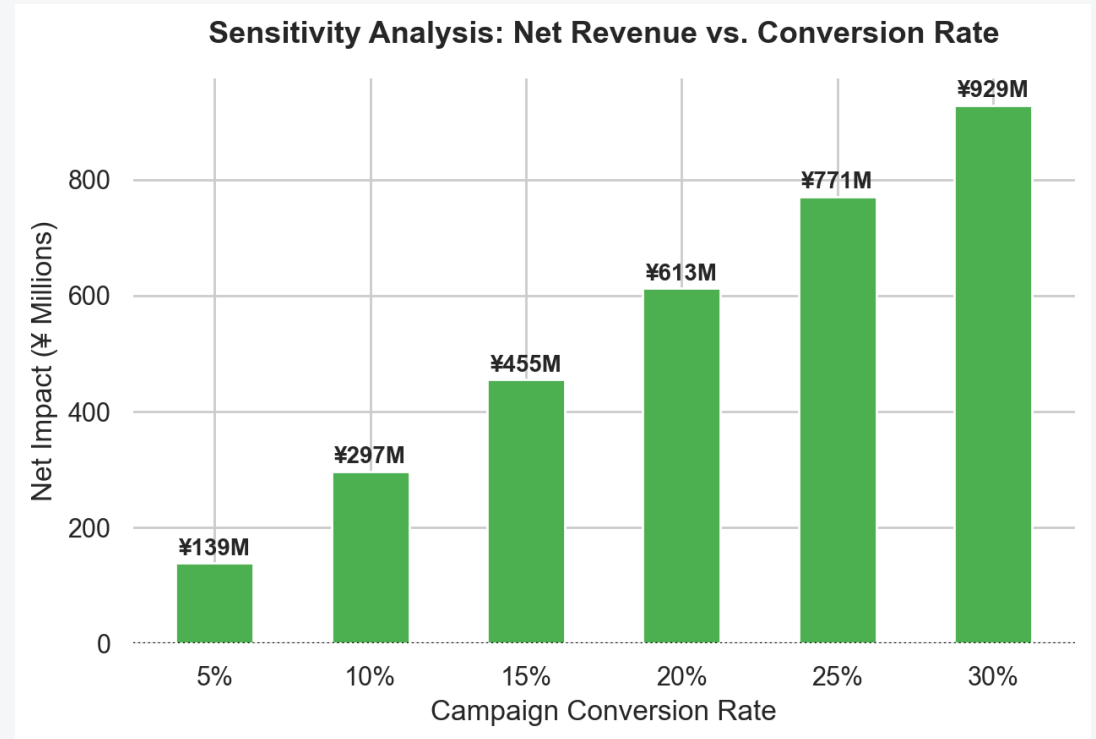
**¥139.7
M**

WORST-CASE NET GAIN

Projected net value at a low 5%
conversion rate

Project Resilience Metrics

- Resilience: Highly profitable even in worst-case scenarios. At a low 5% conversion rate, the campaign generates ¥139.7 Million in net value.
- Brand Protection: Strong precision prevents spamming correctly-aligned users.



COMMERCIAL PROPOSAL & PRICING STRUCTURES

AI Software & Model License (One-Time)

- Model Integration Fee: ¥30,000,000
- Scope of Work:
 - Deployment of the XGBoost pipeline in your local infrastructure.
 - Database connections to automate daily data uploads.
 - Custom dashboard showing top candidates and campaign performance.
- Pricing Justification: This one-time fee corresponds to ~10% of the projected Net Annual Lift of ¥298 Million, showing immediate value.

Long-Term Consulting & Model Updates

- Monthly Retainer Fee: ¥2,500,000 / month
- Scope of Services:
 - Dedicated support from a Senior IT Consultant & Lead Data Scientist.
 - Quarterly model retraining to prevent data drift and capture changing patterns.
 - A/B testing support for marketing campaigns and sales scripts.
- Long-Term Time Series Analysis: Incorporating external economic data and foreign exchange trends to adjust plans.

ROADMAP: EXECUTION TIMELINE

I. Pilot & A/B Testing (Month 1)

- Select a random pilot group of 5,000 flagged customers.
- Contact half with the new offer, keeping the other half as control.
- Validate the assumed 10% conversion rate and ARPU increase.

II. Plan & Script Optimization (Month 2)

- Design the marketing materials and customer portal upgrade screens.
- Train representatives with supportive, 'bill protection' sales scripts.
- Finalize data pipelines.

III. Full-Scale Launch (Month 3+)

- Roll out targeting base-wide to all model-identified customers.
- Monitor retention trends to verify the projected 20% churn decrease.
- Retrain model quarterly.

TECHNICAL DISCLOSURES & REFERENCES

- AI Usage Disclosure: Generative AI (Gemini) was used(Chat history: <https://gemini.google.com/share/3b8fec4b63b6>).
- Project Notebook: The complete codebase is available at mosroomofiz.ipynb, containing all raw charts and feature processing.
- Technical References:
 - Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794. (<https://xgboost.readthedocs.io/>)
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